Ella Standerford

Honors Student in Data Science & Mathematics

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linkedin.com/in/ella-standerford | github.com/ellastanderford | ellastanderford.github.io/portfolio-website/

EDUCATION

Northern Michigan University | Marquette, MI | Expected Graduation: May 2029 | **B.S. in Data Science & Mathematics** | Concentration: Actuarial Science | Certificate in Geographic Information Systems (GIS)| Minor: Business Administration | Honors Program Student | GPA: 4.0 | Dean's List Honoree 2025-present
Negaunee High School | Negaunee, MI | 2021-2025 | Co-Salutatorian | GPA: 4.32

TECHNICAL SKILLS

Data & Analytics: Python (Pandas, Matplotlib), Exploratory Data Analysis (EDA), Statistical Analysis, Preprocessing
Geospatial & Earth Science: Spatial Data Analysis, Pathfinding Algorithms, Environmental Data Modeling
Language & Tools: Java, C, C++, Python, Flask, HTML/CSS, Command Line (Bash)
Business & Professional: Project Management, Technical Documentation, Financial/Business Fundamentals, Public Speaking, Microsoft Office

PROFESSIONAL EXPERIENCE

Analytical Mechanics Associates (AMA) | Hampton, VA
Surface Radiation, Aerosol & Meteorological Measurements Intern | June 2026-December 2026

- Execute exploratory data analysis (EDA) and data cleaning on complex atmospheric radiation and meteorological datasets using Python (Pandas) to support NASA Langley climate research
- Engineer data visualizations (Matplotlib) for deployment on official project webpages
- Extended term through December 2026 to manage additional research data streams

NMU Network Infrastructure Center | Marquette, MI
Student Employee | 2025-Present

- Configure, program, and maintain campus-wide telecommunications
- Manage and analyze service ticket workflows to identify recurring technical issues, streamline route resolution, and maintain IT communication networks across campus

NMU Honors Program | Marquette, MI
Honors Student Ambassador | August 2026-Present

- Deliver presentations to prospective students, leveraging strong communication skills to drive program engagement
- Manage administrative operations, plan campus-wide outreach events, and collaborate across university departments to execute honors initiatives

TECHNICAL PROJECTS

Isle Royale Trail Planner (Python, HTML, CSS)

- Designed a spatial routing tool in Python using graph-theory pathfinding (Dijkstra's Algorithm) to analyze elevation profiles and calculate optimal routes
- Integrated custom geospatial routing logic with a Flask backend and responsive UI to render interactive trail itineraries

Atmospheric Data Visualization Dashboard (Python, Pandas, Matplotlib)

- Built a dashboard to clean, process, and visualize massive environmental and atmospheric datasets.

Interactive Java Maze Game (Java, JavaFX)

- Built a desktop maze game and engineered custom collision logic and event-driven keyboard listeners for real-time player navigation.

Personal Portfolio Website (HTML, CSS, Git)

- Designed and deployed a responsive personal website via GitHub Pages to host professional project documentation

LEADERSHIP & AWARDS

Business Professionals of America (BPA): President (2024-2025), VP (2023-2024). Managed chapter operations; achieved multiple Top 10 National Conference Placements in Microsoft Events.
Key Club and NHS: Key Club President (2024-2025); National Honor Society Treasurer (2024-2025).
NMU Campus Engagement: Participant in Student Leader Fellowship Program (SLFP), Superior Edge, Quiz Bowl, Letters of Love, the Honors Program, and treasurer of the Honors Student Organization.
Award: National Center for Women & Information Technology (NCWIT) Regional Affiliate Award (2025).

RELEVANT COURSEWORK

Calculus I & II, Linear Algebra, Discrete Mathematics, Computer Science I & II, Programming in C++